Lecture 3

Monoidal categories and string diagrams

Objectives. *Lecture 2 showed that a semisimple category alone cannot tell* $\mathrm{Rep}(S_3)$ *from* $\mathrm{Rep}(\mathbb{Z}_3)$*: what can is the tensor product. This lecture adds it. Putting systems side by side is a functor* $\otimes$ *with an associator and unitors, subject to two axioms, the pentagon and the triangle. Coherence says that these axioms are all one needs, and string diagrams turn it into a calculus of pictures. Associators need not be trivial: on* $\mathbb{Z}_2$ *there are two monoidal categories with the same fusion rules and different associators.*

By the end of this lecture you will be able to:

- State the definition of a monoidal category, check it on Vect, $\mathrm{Rep}(G)$ and the matrices, and draw the pentagon and the triangle.
- Read and draw string diagrams, prove the interchange law, and use sliding and the commutativity of scalars in pictures.
- Show that the pentagon of Vec_G^ω is the 3-cocycle condition, and solve it for $\mathbb{Z}_2$.
- Define a monoidal functor, and show that $\omega(s,s,s) = \pm 1$ cannot be changed by one that is the identity on objects.

3.1 The definition

For objects of Vect, $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$ are different spaces, identified by $(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$. A monoidal category keeps track of such identifications.

Definition 3.1 (Monoidal category)**.** A *monoidal category* is a category C with

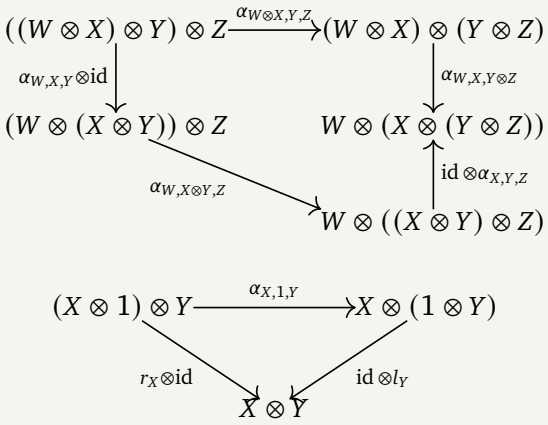
- a *tensor product*: an object $X \otimes Y$ for all objects and a morphism $f \otimes g : X \otimes Y \to X' \otimes Y'$ for all $f : X \to X'$ and $g : Y \to Y'$, with $\mathrm{id}_X \otimes \mathrm{id}_Y = \mathrm{id}_{X \otimes Y}$ and

$$\begin{aligned}(g \otimes k) \circ (f \otimes h) \\ = (g \circ f) \otimes (k \circ h);\end{aligned}$$

- a *unit object* 1;
- natural isomorphisms, the *associator* and the *unitors*,

$$\begin{aligned}\alpha_{X,Y,Z} &: (X \otimes Y) \otimes Z \\ &\to X \otimes (Y \otimes Z), \\ l_X &: 1 \otimes X \to X, \\ r_X &: X \otimes 1 \to X,\end{aligned}$$

such that the *pentagon* and the *triangle* commute for all objects:



It is *strict* if α, l and r are identities.

The first condition says that $\otimes$ is a functor of two variables; we return to it in § 3.3.

Read the pentagon from the upper-left object to the middle-right object. The upper route first groups Y with Z, then groups their pair with X. The lower route first groups X with Y, then regroups that pair with Z, splitting off W, then groups Y with Z inside the last three factors. Each arrow changes brackets only. The order W, X, Y, Z stays fixed. The axiom says that these two composites, with two and three arrows, agree.

Example 3.2 (Monoidal categories)**.**

- $(\mathrm{Vect}, \otimes, \mathbb{C})$, the state spaces of composite quantum systems (*QM-I* Post. 10.1), with α as above and $l(z \otimes v) = zv = r(v \otimes z)$. Both axioms hold on pure tensors, which span: in the pentagon both paths send

$$\begin{aligned} &((w \otimes x) \otimes y) \otimes z \\ &\mapsto w \otimes (x \otimes (y \otimes z)). \end{aligned}$$

  (*Linear Algebra* Rem. 12.6 writes the inverse of our α.)
- $(\mathrm{Set}, \times, \{*\})$, with $\alpha((x, y), z) = (x, (y, z))$.
- $\mathrm{Rep}(G)$: g acts on $V \otimes W$ by $\rho_V(g) \otimes \rho_W(g)$, a representation by the first condition; the unit is the trivial representation $\mathbb{C}$, and α, l, r of Vect are intertwiners. For SU(2) two spin-$\frac{1}{2}$ representations give the singlet plus the triplet, $\Pi_{1/2} \otimes \Pi_{1/2} \cong \Pi_0 \oplus \Pi_1$ (*Symmetries* Thm. 14.12).
- $\mathrm{Mat}_{\mathbb{C}}$ (Lecture 1) with $m \otimes n = mn$ and the Kronecker product of matrices: it is *strict*, because

$$(A \otimes B) \otimes C = A \otimes (B \otimes C),$$
$$1 \otimes A = A = A \otimes 1$$

  hold as equalities of matrices. The quantum circuits of *Quantum Computing* compute in it. They are drawn left to right with the top wire as the first factor; turned a quarter turn anticlockwise they become the pictures of this course, with time upward and the first factor on the left.

3.2 Coherence (formal diagrams)

In $W \otimes X \otimes Y \otimes Z$ there are five bracketings and many composites of associators between them. The pentagon equates two of them; Mac Lane's theorem says this is enough.

Theorem 3.3 (Coherence · *cited*).

(a) Every monoidal category is monoidally equivalent (§ 3.6) to a strict one.

(b) Take a word $x_1 \otimes \cdots \otimes x_n$ in formal variables. Any two formal *morphisms between two of its bracketings, built from α, l, r, their inverses and identities by $\circ$ and $\otimes$, are equal, and stay equal when objects of C are put in for the variables.*

(Mac Lane, §VII.2 and §XI.3; EGNO Thms. 2.8.5 and 2.9.2.)

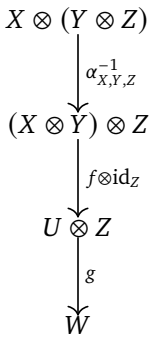
What must be formal is the morphism, not just its source and target. Mac Lane points out that two formally different bracketings may become the same object, and then a morphism between them need not be formal. The category Vec_G^ω on $\mathbb{Z}_2$ has simple objects 1 and s, with $s \otimes s = 1$ (§ 3.5). For $\omega(s,s,s) = -1$,

$$(s \otimes s) \otimes s = s = s \otimes (s \otimes s),$$

yet $\alpha_{s,s,s} = -\mathrm{id}_s$: the diagram formed by $\alpha_{s,s,s}$ and id_s does not commute. There is no contradiction: id_s compares the two bracketings only because they happen to be the same object, and it is not a formal morphism from $(x \otimes x) \otimes x$ to $x \otimes (x \otimes x)$. Some textbooks state coherence as "every diagram commutes"; that is false.

What (a) and (b) give us: in a formal computation the brackets and the associators can be dropped, and we write $X \otimes Y \otimes Z$.

Dropping brackets is a convention for writing composites, not an assertion that the chosen category has identity associators. For example, let $f\colon X \otimes Y \to U$ and $g\colon U \otimes Z \to W$. The picture that applies f to the first two strands and then applies g has the following interpretation if its source is bracketed as $X \otimes (Y \otimes Z)$:



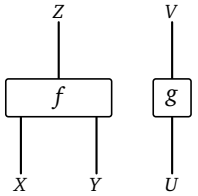
In a strict category the first arrow is an identity. In a skeletal category such as Vec_G^ω, the first arrow can be a nontrivial scalar even when its source and target are the same object. One must insert that scalar when evaluating the picture in that model. Strictification supplies an equivalent category; it does not replace $\alpha_{s,s,s} = -\mathrm{id}_s$ by an identity inside the original category.

3.3 String diagrams; interchange; Joyal–Street

In a strict monoidal category the pictures of Lecture 1 acquire a second direction. A morphism $f\colon X_1 \otimes \cdots \otimes X_m \to Y_1 \otimes \cdots \otimes Y_n$ is a box with m strands entering from below and n leaving

above, the first factor on the left. Composition stacks boxes upward, $\otimes$ places pictures side by side, id_X is a strand, and the unit 1 is the empty picture: it has no strand. For $f\colon X \otimes Y \to Z$ and $g\colon U \to V$,

$$f \otimes g \;=\; \tag{3.1}$$



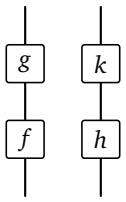
a morphism $X \otimes Y \otimes U \to Z \otimes V$.

Theorem 3.4 (Interchange). *For composable f, g and composable h, k,*

$$\begin{aligned}(g \otimes k) \circ (f \otimes h)\\ = (g \circ f) \otimes (k \circ h).\end{aligned}$$

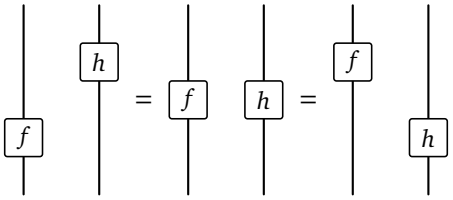
Proof. This is the first condition of Definition 3.1: $\otimes$ is a functor on pairs of morphisms, which compose slot by slot. ∎

The theorem is what makes a picture unambiguous. The diagram



$= (g \otimes k) \circ (f \otimes h)$ (3.2)

can be read row by row or column by column, and the two readings agree. As a special case, boxes on different strands slide past each other:



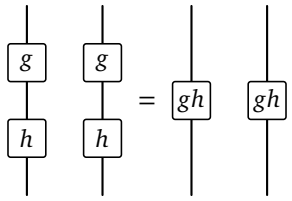
(3.3)

that is,

$$\begin{aligned}(\mathrm{id} \otimes h) \circ (f \otimes \mathrm{id}) &= f \otimes h\\ &= (f \otimes \mathrm{id}) \circ (\mathrm{id} \otimes h).\end{aligned}$$

The heights of boxes on different strands carry no meaning.

A first proof by pictures: the action of G on $V \otimes W$ in Example 3.2 is a representation. Writing g for $\rho_V(g)$ and for $\rho_W(g)$, interchange and $\rho(g)\rho(h) = \rho(gh)$ on each strand give



(3.4)

that is,

$$\begin{aligned}(\rho_V(g) \otimes \rho_W(g))(\rho_V(h) \otimes \rho_W(h))\\ = \rho_V(gh) \otimes \rho_W(gh).\end{aligned}$$

Theorem 3.5 (Joyal–Street · *cited*)**.** *In a strict monoidal category, two string diagrams that can be deformed into one another in the plane, keeping the ends of the strands fixed, never letting strands cross, and never letting a strand turn downward, denote the same morphism (Heunen–Vicary Thm. 1.8; Selinger Thm. 3.1).*

By coherence (Theorem 3.3) the theorem applies to any monoidal category: a picture determines the morphism once its source and target are bracketed, and by (b) every way of inserting associators to match them gives the same result. *Quantum Computing*'s soundness theorem for circuits (L2, Thm. 2.2) is the case of $\mathrm{Mat}_{\mathbb{C}}$, read a quarter turn round.

3.4 Eckmann–Hilton

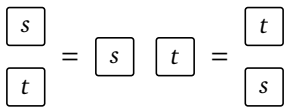
The endomorphisms of the unit, $\mathrm{End}(1)$, are the *scalars* of the category. In Vect, $\mathrm{End}(\mathbb{C}) = \mathbb{C}$. In a picture a scalar is a box with no strands.

Theorem 3.6 (Scalars commute · *proved given coherence*)**.** *In a monoidal category, $s \circ t = t \circ s$ for all $s, t \in \mathrm{End}(1)$.*

Proof. By Theorem 3.3(a) it suffices to take C strict, so that $1 \otimes 1 = 1$ and $s \otimes \mathrm{id}_1 = s = \mathrm{id}_1 \otimes s$. Then, by interchange,

$$\begin{aligned} s \circ t &= (s \otimes \mathrm{id}_1) \circ (\mathrm{id}_1 \otimes t) \\ &= s \otimes t \\ &= (\mathrm{id}_1 \otimes t) \circ (s \otimes \mathrm{id}_1) = t \circ s. \end{aligned}$$

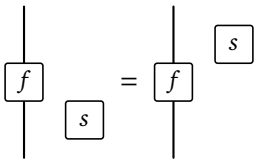
In pictures, two boxes with no strands slide past each other:



(3.5)

■

A box with no strands can slide up and down beside a strand, on the same side of it,



(3.6)

Moving it to the other side of the strand is not a planar deformation, and in a general monoidal category $s \otimes \mathrm{id}_X$ and $\mathrm{id}_X \otimes s$ can differ (Heunen–Vicary Ex. 1.9). In Vect, and in the fusion categories of Lecture 5, whose tensor product is bilinear, a scalar is a number, $s \otimes f = sf = f \otimes s$, and it floats anywhere.

3.5 Vec_G^ω: the $\mathbb{Z}_2$ pentagon

The associator is data. The simplest place to see that it can be non-trivial is a category whose simple objects form a group.

Definition 3.7 (Vec_G^ω)**.** Let G be a finite group and $\omega : G^3 \to \mathbb{C}^\times$ a function. Vec_G^ω is the skeletal semisimple category with one simple object δ_g for each $g \in G$, tensor product

$\delta_g \otimes \delta_h = \delta_{gh}$, unit δ_1, and associator

$$\alpha_{g,h,k} = \omega(g,h,k)\ \mathrm{id}_{\delta_{ghk}}$$

on simple objects, extended to direct sums by linearity, with the tensor product of linear maps on morphisms; the unitors are identities. We write g for δ_g. $\mathrm{Vec}_G = \mathrm{Vec}_G^1$ is equivalent to the category of finite-dimensional G-graded vector spaces. Here ω will be a cocycle, not a symplectic form, a quantum state, or the root of unity of Lecture 2.

Naturality is automatic, since the only morphisms between simple objects are scalars. The two axioms constrain ω.

Theorem 3.8 (The pentagon is a cocycle condition)**.** *Vec_G^ω is a monoidal category if and only if*

$$\omega(g,1,h) = 1$$

for all g,h (the triangle; ω is normalised*) and, for all $g,h,k,m \in G$,*

$$\begin{aligned}\omega(gh,k,m)\,\omega(g,h,km)\\ = \omega(g,h,k)\,\omega(g,hk,m)\\ \cdot\,\omega(h,k,m)\end{aligned}$$

(the pentagon; ω is a 3-cocycle*).*

Proof. By linearity it suffices to check the axioms on simple objects, where every arrow is a scalar times an identity and $\otimes\,\mathrm{id}$ does not change the scalar. In the pentagon for g,h,k,m the two-arrow path is $\omega(gh,k,m)\,\omega(g,h,km)$, and the three-arrow path is $\omega(g,h,k)\,\omega(g,hk,m)\,\omega(h,k,m)$. In the triangle for g,h, the path through $\alpha_{g,1,h}$ is $\omega(g,1,h)$, and the other is 1. ∎

Example 3.9 ($\mathbb{Z}_2$)**.** Let $G = \mathbb{Z}_2 = \{1,s\}$. Substituting $h = 1$ in the cocycle condition and using $\omega(g,1,k) = 1$ gives

$$\omega(g,k,m) = \omega(g,k,m)\,\omega(1,k,m).$$

Since the values are nonzero, $\omega(1,k,m) = 1$. Substituting $k = 1$ likewise gives $\omega(g,h,1) = 1$. Only $\omega(s,s,s)$ is free. Of the sixteen pentagons only (s,s,s,s) is not automatic, and since $s^2 = 1$ it reads

$$1 = \omega(s,s,s)^2.$$

So $\omega(s,s,s) = \pm 1$: there are two monoidal categories Vec_G^ω on $\mathbb{Z}_2$, with the same fusion rule $s \otimes s = 1$.

3.6 Monoidal functors

Are the two categories of Example 3.9 really different, or the same category described twice? To compare monoidal categories we need functors that respect $\otimes$.

Definition 3.10 (Monoidal functor)**.** A *monoidal functor* $(F, J, J_0) : C \to \mathcal{D}$ is a functor F with natural isomorphisms

$$J_{X,Y} : F(X) \otimes F(Y) \to F(X \otimes Y),$$
$$J_0 : 1 \to F(1),$$

such that

$$\begin{aligned} &F(\alpha_{X,Y,Z}) \circ J_{X\otimes Y,Z} \\ &\quad \circ (J_{X,Y} \otimes \mathrm{id}) \\ &= J_{X,Y\otimes Z} \circ (\mathrm{id} \otimes J_{Y,Z}) \\ &\quad \circ \alpha_{FX,FY,FZ} \end{aligned}$$

and

$$F(l_X) \circ J_{1,X} \circ (J_0 \otimes \mathrm{id}) = l_{F(X)},$$
$$F(r_X) \circ J_{X,1} \circ (\mathrm{id} \otimes J_0) = r_{F(X)}.$$

It is a *monoidal equivalence* if F is an equivalence. Between $\mathbb{C}$-linear categories we take F to be $\mathbb{C}$-linear on every Hom space, as all functors are from here on.

The forgetful functor $\mathrm{Rep}(G) \to \mathrm{Vect}_{\mathrm{fd}}$, with J and J_0 identities, is monoidal.

The tensor product already answers the question left open in Lecture 2. $\mathrm{Rep}(\mathbb{Z}_3)$ has the fusion rules of Vec_G for $G = \mathbb{Z}_3$, since its characters multiply, $\chi_i \otimes \chi_j = \chi_{i+j}$: every simple object is invertible. In $\mathrm{Rep}(S_3)$ the standard representation is not: with the notation of Lecture 2,

$$\mathrm{std} \otimes \mathrm{std} \cong \mathbf{1} \oplus \mathrm{sgn} \oplus \mathrm{std},$$

of dimension $4 = 1 + 1 + 2$. So the two categories, equivalent as linear categories, are not equivalent as monoidal ones: a monoidal equivalence would send simple objects to simple objects and respect their tensor products.

Back to Example 3.9. For a functor $\mathrm{Vec}_G^{\omega} \to \mathrm{Vec}_G^{\omega'}$ that is the identity on objects, with $J_{g,h} = j(g,h)\,\mathrm{id}$ and $J_0 = j_0\,\mathrm{id}$, the first condition reads

$$\begin{aligned} &\omega'(g,h,k)\, j(h,k)\, j(g,hk) \\ &\quad = j(g,h)\, j(gh,k)\, \omega(g,h,k), \end{aligned}$$

and the two unit conditions, with trivial unitors, read $j(1,g) = j(g,1) = j_0^{-1}$.

For a general j the first condition says $\omega' = \omega\,(\delta j)^{-1}$, where the *coboundary* of j is

$$(\delta j)(g,h,k) = \frac{j(h,k)\, j(g,hk)}{j(gh,k)\, j(g,h)}.$$

If ω and ω' are both normalised, the condition at $(g,1,h)$ gives $j(g,1) = j(1,h)$, so such a j satisfies the unit conditions with $j_0 = j(1,1)^{-1}$. Normalised cocycles that differ by a coboundary therefore give equivalent categories. Their classes form $H^3(G,\mathbb{C}^\times)$, the 3-cocycles modulo coboundaries. Every class contains a normalised cocycle (EGNO Rem. 2.6.3); that is why Example 3.9 could normalise from the start.

Proposition 3.11. *A monoidal functor $\mathrm{Vec}_G^{\omega} \to \mathrm{Vec}_G^{\omega'}$ on $\mathbb{Z}_2$ that is the identity on objects exists only if $\omega'(s,s,s) = \omega(s,s,s)$.*

Proof. At $(g,h,k) = (s,s,s)$, with $s^2 = 1$, the first condition reads

$$\begin{aligned} &\omega'(s,s,s)\, j(s,s)\, j(s,1) \\ &\quad = j(s,s)\, j(1,s)\, \omega(s,s,s), \end{aligned}$$

and $j(s,1) = j(1,s)$ by the unit conditions, or, for normalised cocycles, by the condition at $(s,1,s)$. ∎

By EGNO Prop. 2.6.1 · *cited*, two categories Vec_G^{ω} and $\mathrm{Vec}_G^{\omega'}$ are equivalent by a $\mathbb{C}$-linear monoidal functor exactly when ω' is obtained from ω by an automorphism of G and a coboundary; their classes form $H^3(G, \mathbb{C}^\times)$ up to automorphisms of G. The only automorphism of $\mathbb{Z}_2$ is the identity, and Proposition 3.11 shows that no coboundary between normalised cocycles changes $\omega(s,s,s)$. So the two categories of Example 3.9 are not equivalent: the same fusion rules carry two different associators. Lecture 6 computes associators as F-symbols, and for Vec_G^{ω} they are the values of ω; Lecture 7 finds two braidings for each.

Summary.

- *Handed on:* a monoidal category is $(C, \otimes, 1, \alpha, l, r)$ with the pentagon and the triangle (Definition 3.1); by coherence, formal diagrams commute and brackets can be dropped (Theorem 3.3), but non-formal ones need not.
- *Pictures:* boxes and strands with time upward; interchange makes them unambiguous, boxes on different strands slide, and planar deformations preserve the morphism (Theorems 3.4 and 3.5).
- *Scalars commute* (Theorem 3.6).
- *The main example:* in Vec_G^{ω} the pentagon is the 3-cocycle condition; on $\mathbb{Z}_2$, $\omega(s,s,s) = \pm 1$, and no monoidal functor that is the identity on objects relates the two (Theorem 3.8 and Proposition 3.11).
- *Next:* Lecture 4 adds duals: cups, caps and loops.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: monoidal categories, Def. 2.1.1, p. 21, and Def. 2.2.8, p. 25; Vec_G^{ω}, Ex. 2.3.8, p. 28; monoidal functors, Def. 2.4.1, p. 30; End(1) commutative, Prop. 2.2.10, p. 26; strictification, Thm. 2.8.5, p. 36; coherence, Thm. 2.9.2, p. 40; the classification, Prop. 2.6.1 and Rem. 2.6.3, p. 33. Mac Lane, *Categories for the Working Mathematician*: §VII.1–2, pp. 161–169 ("formal" diagrams, p. 165; his associator runs the other way), and §XI.3, p. 257. Heunen and Vicary, *Categories for Quantum Theory*: Def. 1.1, p. 30; interchange, Thm. 1.7, p. 32; the graphical calculus, Thm. 1.8, p. 34; boxes cannot pass strands, Ex. 1.9, p. 35; scalars, Lemma 2.3, p. 62. Selinger, *A survey of graphical languages for monoidal categories*, arXiv:0908.3347, §3, Thm. 3.1, p. 11 (drawn left to right).